ADMINISTRATIVE NOTICE

NOT MEASUREMENT SENSITIVE

JOINT-UXS-002
NOTICE 2
26 August 2026

SUPERSEDING
NOTICE 1
20 January 2022

MODULAR OPEN SYSTEMS APPROACH (MOSA) PORTFOLIO – HARDWARE OPEN SYSTEMS TECHNOLOGIES (HOST), FUTURE AIRBORNE CAPABILITY ENVIRONMENT® OR FACE®, SENSOR OPEN SYSTEMS ARCHITECTURE® OR SOSA®, UNMANNED SYSTEMS (UXS) CONTROL SEGMENT (UCS), AND OPEN MISSION SYSTEMS (OMS)

MODULAR OPEN SYSTEMS APPROACH (MOSA) – STANDARDS ADOPTION

The U.S. Navy, U.S. Air Force, and U.S. Army have continued to push forward with the development and adoption of MOSA. Mature Open Architecture (OA) standards, and alignment between standards, has allowed airborne systems to maximize the benefits of MOSA. This MOSA portfolio supports airborne systems, with many successful implementations of unmanned systems. The following five MOSA standards demonstrate applications from a low level viewpoint to higher level viewpoints: Hardware Open Systems Technologies (HOST), Future Airborne Capability Environment or FACE, Sensor Open Systems Architecture or SOSA, Unmanned Systems (UxS) Control Segment (UCS), and Open Mission Systems (OMS).

STANDARD SPECIFICATION

HARDWARE OPEN SYSTEMS TECHNOLOGIES (HOST)

HOST is a standards framework that defines multiple OAs that support a MOSA and standardizes the implementation of Commercial-Off-the-Shelf (COTS) components for embedded computing on open systems for U.S. defense platforms.

HOST enables the development of interoperable, modular, and upgradeable systems across military platforms and leverages open standards across product families to improve defense acquisition.

More information regarding the HOST standards framework can be found via https://www.navair.navy.mil/host/.

STANDARD SPECIFICATION

FUTURE AIRBORNE CAPABILITY ENVIRONMENT – THE OPEN GROUP FACE CONSORTIUM

The Open Group FACE Consortium standards are now referenced within ASSIST via:

FACE-ED.3.2 - Future Airborne Capability Environment (FACE™) Ed. 3.2

More information regarding the FACE technical standard can be found via https://publications.opengroup.org/standards/face.

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STANDARD SPECIFICATION

SENSOR OPEN SYSTEMS ARCHITECTURE

The Open Group SOSA Consortium standards are now referenced within ASSIST via:

SOSA-ED.2.0-SNAP3 - Sensor Open Systems Architecture (SOSA™) Ed. 2.0 (Snapshot 3)

More information regarding the SOSA aligned/upcoming technical standard can be found via https://publications.opengroup.org/standards/sosa.

STANDARD SPECIFICATION

UNMANNED SYSTEMS (UxS) CONTROL SEGMENT (UCS) – SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

The UCS architecture was started out of the Office of Secretary of Defense (OSD) Acquisition, Technology, and Logistics (AT&L), mandated across the Tri-Services for all Unmanned Aircraft Systems (UAS), and later adopted by the Navy for all UxS within its portfolio. The UCS standard was developed with DoD services and industry partners and transitioned to the SAE AS-4 UxS, Aerospace Standard (AS) working group for continued collaboration across all UxS stakeholders. The UCS architecture provides a commercially available and used Unified Modeling Language (UML) as a Service Oriented Architecture (SOA), which includes a data dictionary, a modeling framework, a specification, integration capabilities, and implementation for deployment of all UxS software across the Tri-Services. The UCS open standard architecture has the following goals:

- Enabling small business competition
- Increased capabilities while achieving rapid development and cost reduction across development, sustainment, and life-cycle management
- Some platform independence, which enables UCS architecture to be leveraged across different computing infrastructures and different communication protocols (e.g., Air Forces OMS, Joint Architecture for Unmanned Systems [JAUS], FACE, and Joint All-Domain Command and Control [JADC2])
- Successful reuse of software and commonality across Tri-Service platforms

UCS technical standards, based on SAE AS6518A, can be found via https://www.sae.org/standards/content/as6518a/.

STANDARD SPECIFICATION

OPEN MISSION SYSTEMS (OMS)

The goal of the OMS is to develop industry and DoD consensus for a non-proprietary mission system architectural standard that enables affordable technical refresh and insertion, simplified mission systems integration, service reuse and interoperability, and competition across the life cycle. Industry and the Government have been working on solutions cooperatively to achieve these goals for several years with extensive testing and demonstrations.

The OMS standard can be found via https://dle.afrl.af.mil and https://gitlab.com/open-arsenal/oms.

OTHER STANDARDS AND PATH FORWARD

The standards here are a breadth of today's leveraged standards. The Navy will continue on its path by leading the way in reviewing, developing, and participating in future standards as it pertains to benefiting rapid development, cost reduction, and interoperability across its portfolio with new capabilities for tomorrow's warfighter.

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CONCLUDING MATERIAL

Custodians:
Army – OCSE
Navy – SH
Air Force – 10
OSD – SE

Review activities:
Army – MI
Navy – AS, CG, CH, EC, MC, NM, OS, SA
Air Force – 01
DLA – CC

Preparing activity:
Navy – SH
(Project MOSS-2026-014)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using ASSIST at https://assist.dla.mil.

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